Quantification: 542_z06
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	15
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2576	3013	1380.42	px^2
major_axis_length_px	lacuna shape	109.567	124.464	38.2072	px
minor_axis_length_px	lacuna shape	31.8891	33.1799	10.1914	px
aspect_ratio	lacuna shape	3.6142	3.6285	1.2579	unitless
eccentricity	lacuna shape	0.9409	0.9613	0.0498	unitless
solidity	lacuna shape	0.7493	0.7672	0.1286	unitless
roots_count	headline	14.5333	15	6.7809	count
ring_length_r30_px	headline	473.2	423	202.636	px
ring_length_r60_px	reported	1032.73	981	449.129	px
owned_length_px	ownership-dependent	2075.13	2042.07	1418.74	px
edge_count	ownership-dependent	93.3333	85	65.4181	count
mean_edge_length_px	ownership-dependent	22.3671	21.9229	3.3454	px
perimeter_px	normalised	292.07	309.522	122.808	px
ring_area_r30_px2	normalised	10491.5	11337	3304.03	px^2
ring_area_r60_px2	normalised	24400.9	26961	7815.16	px^2
in_frame_fraction_r30	normalised	1	1	0.0001	unitless
in_frame_fraction_r60	normalised	0.9858	1	0.0377	unitless
ring_density_r30	normalised	0.0440387	0.0446422	0.00754502	px^-1
ring_density_r60	normalised	0.0410994	0.040433	0.00712237	px^-1
roots_per_100px_perimeter	normalised	5.1783	4.8462	1.5065	per 100 px
ring_attached_length_r30_px	network v2	424.867	364	203.634	px
ring_attached_length_r60_px	network v2	852.933	731	472.517	px
ring_length_w_r30_px	network v2	538.17	477.676	233.742	px
ring_length_w_r60_px	network v2	1172.36	1098.92	516.356	px
sholl_crossings_r10	network v2	10.7333	10	5.7998	count
sholl_crossings_r20	network v2	13.1333	14	5.4362	count
sholl_crossings_r30	network v2	13.6	12	5.4485	count
ring_width_mean_r30_px	width	7.4697	7.5343	0.4396	px

Per field
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measure	value	units
analysed_area_px2	1.0064e+06	px^2
total_skeleton_length_px	39890	px
canalicular_length_density_per_px	0.0396362	px^-1
junction_count	905	count
junction_density_per_px2	0.00089924	px^-2
lacuna_count	19	count
lacunae_per_px2	1.888e-05	px^-2
skeleton_component_count	235	count

mean_component_length_px	169.745	px
median_component_length_px	26	px
field_length_density_w_per_px	0.0448956	px^-1
field_density_without_flagged_per_px	0.0396045	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	375	px
unattached_component_count	20	count
canalicular_length_density_attached_per_px	0.0396362	px^-1
field_length_density_with_unattached_per_px	0.0400088	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	8.1721	px
width_median_px	8.2462	px
width_p10_px	4.4721	px
width_p90_px	10.7703	px
width_pixels_used	38643	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1381	54.5303	43.8441	1.2437	0.5946	0.657	-0.6304	16.7227	909.274	True
2	445	56.9837	11.5357	4.9398	0.9793	0.7672	-0.2356	148.654	554.791	False
3	1001	62.1533	28.1242	2.21	0.8918	0.5533	-0.1968	55.7832	107.949	False
4	895	46.7108	27.3564	1.7075	0.8106	0.7956	0.2016	143.341	635.06	False
5	2450	125.59	27.3191	4.5971	0.9761	0.6618	-0.1995	179.65	380.673	False
6	4645	128.217	49.4579	2.5925	0.9226	0.72	-0.0979	201.844	870.357	False
7	2177	80.2406	35.2894	2.2738	0.8981	0.8959	-0.0663	304.003	624.629	False
8	3251	124.464	40.9538	3.0391	0.9443	0.639	0.028	423.426	66.5638	False
9	3239	125.767	33.4856	3.7559	0.9639	0.9114	-0.0886	481.731	573.277	False
10	3188	122.408	33.7355	3.6285	0.9613	0.8804	-0.0452	529.624	391.03	False
11	3013	170.645	29.6128	5.7626	0.9848	0.5248	-0.2083	594.958	912.812	False
12	3414	148.999	33.1799	4.4906	0.9749	0.8294	-0.0805	580.581	782.628	False
13	4647	139.214	45.6902	3.0469	0.9446	0.8623	-0.0076	723.033	247.267	False
14	1123	92.9757	18.8182	4.9407	0.9793	0.6997	-0.1287	817.288	724.433	False
15	3750	145.258	33.093	4.3894	0.9737	0.8965	-0.0485	952.204	400.276	True
16	1341	90.1135	19.5596	4.6071	0.9762	0.911	-0.0956	984.752	948.262	True
17								310.754	226.539	False
18								933.644	3.9103	True
19								368.378	576.748	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	8	224	587	964.453	35	27.5558	7.8052
2	5	210	518	843.269	33	25.5536	7.6954
3	14	390	709	1477.15	54	27.3547	7.5343
4	11	286	688	673.044	27	24.9275	6.8995
5	10	413	1064	2237.97	96	23.3122	7.2249

6	28	837	1714	5172.17	219	23.6172	7.5889
7	16	423	762	832.784	38	21.9154	7.3911
8	16	482	906	966.009	47	20.5534	7.6519
9	8	406	981	2042.07	94	21.7242	8.5432
10	15	512	1310	4028.42	162	24.8668	7.6136
11	20	815	1834	4102.75	220	18.6489	7.3385
12	22	652	1393	2361.16	115	20.5318	7.0766
13	23	636	1349	2398.68	127	18.8873	7.7579
14	10	300	702	2268.77	85	26.6914	6.5975
15	8	372	904	2422.03	92	26.3264	7.5598
16	4	154	505	65.4558	6	10.9093	5.3984
17	16	582	1325	1512.68	69	21.9229	7.6131
18	2	77	197	399.191	18	22.1773	8.425
19	4	154	236	209.983	14	14.9988	7.5189

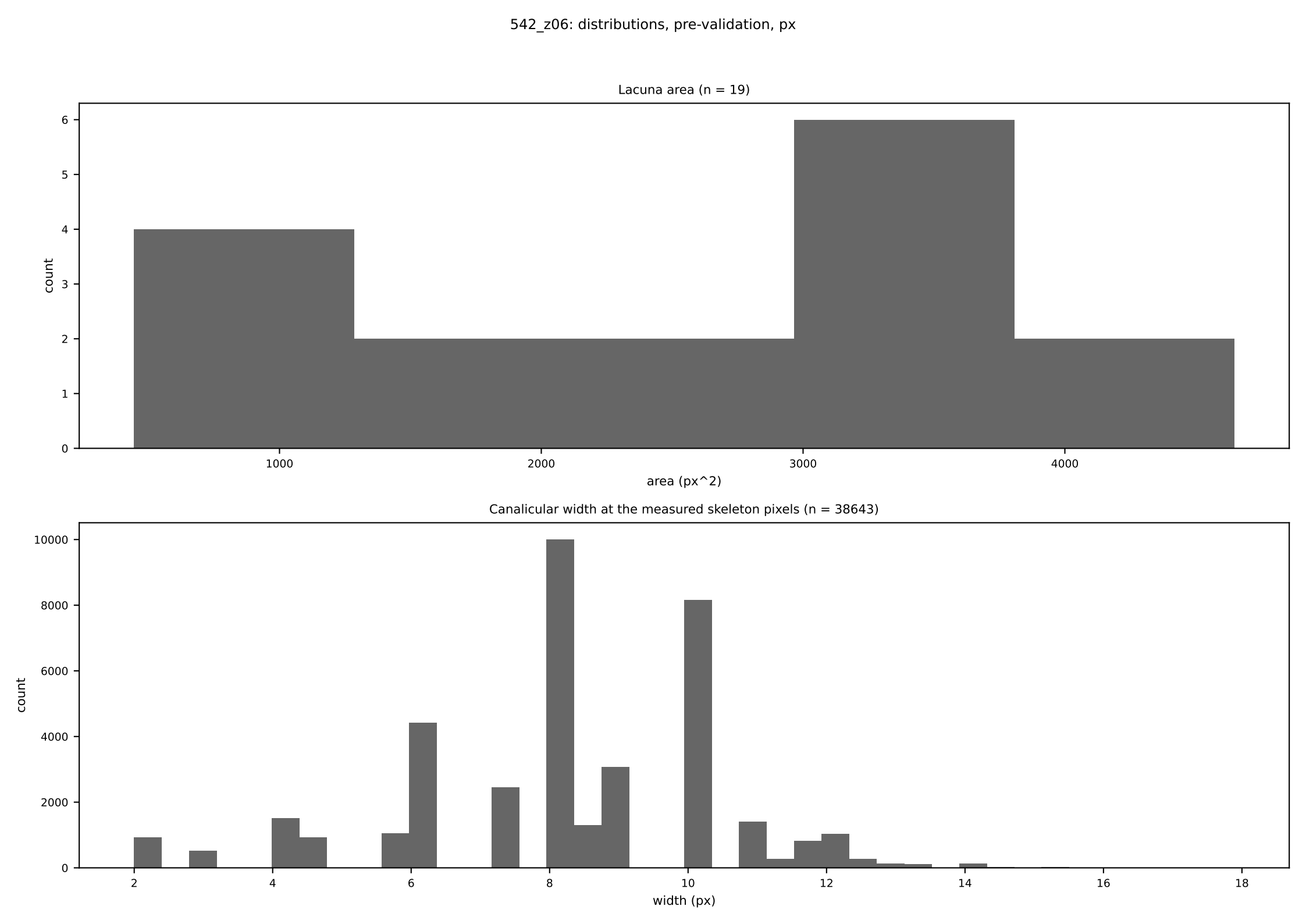
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	215.232	5617	12694	0.6363	0.5983	0.0398789	0.0462423	3.7169
2	126.468	6226	14794	1	1	0.0337295	0.0350142	3.9536
3	225.309	8379	19327	0.9998	0.884	0.0465449	0.0366844	6.2137
4	151.782	6752	15321	1	1	0.0423578	0.0449057	7.2472
5	322.043	11663	28685	1	1	0.0354111	0.0370926	3.1052
6	438.517	13901	31616	1	1	0.0602115	0.0542131	6.3852
7	211.924	8423	18846	1	1	0.0502196	0.040433	7.5499
8	359.191	12383	26961	1	0.9025	0.0389243	0.0336041	4.4545
9	302.451	10940	24775	1	1	0.0371115	0.0395964	2.6451
10	309.522	11337	28286	1	1	0.0451619	0.0463127	4.8462
11	468.274	15363	34096	1	1	0.0530495	0.0537893	4.271
12	422.978	13165	30001	1	1	0.0495253	0.0464318	5.2012
13	379.848	13296	32108	1	1	0.0478339	0.0420145	6.0551
14	217.924	8830	22986	1	1	0.0339751	0.0305403	4.5888
15	332.894	10505	23806	0.8602	0.7919	0.0354117	0.0379736	2.4032
16	205.776	6766	16357	0.7841	0.714	0.0227609	0.0308736	1.9439
17	387.948	13037	31120	1	1	0.0446422	0.0425771	4.1243
18	87.0061	2915	8653	0.5433	0.5282	0.0264151	0.0227667	2.2987
19	56.8701	3677	7091	1	1	0.041882	0.0332816	7.0336

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	224	580	259.137	671.772	6	5	5
2	196	392	242.38	583.76	5	8	10
3	330	654	445.576	803.826	8	7	10
4	241	368	319.037	769.328	5	7	8
5	334	859	475.789	1217.67	9	14	13
6	834	1619	964.921	1971.29	24	20	22
7	364	560	477.676	862.926	8	12	11
8	453	731	543.446	1016.15	12	11	10
9	314	550	456.747	1098.92	11	14	12
10	501	1290	589.144	1504.98	11	14	16
11	715	1723	930.696	2072.23	15	22	25
12	610	1227	732.257	1573.77	18	16	15
13	611	1259	720.571	1527.18	17	22	19
14	259	579	334.037	789.227	6	9	9
15	310	705	418.605	1017.39	9	11	16
16	119	119	170.953	558.546	4	7	6

17	503	859	672.47	1535.91	10	16	18
18	61	156	89.0122	222.51	2	3	4
19	108	124	167.811	258.166	2	5	6


542_z06: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
1
2
3
4
5
6
1000
2000
3000
4000
Canalicular width at the measured skeleton pixels (n = 38643)
count
width (px)
0
2000
4000
6000
8000
10000
2
4
6
8
10
12
14
16
18